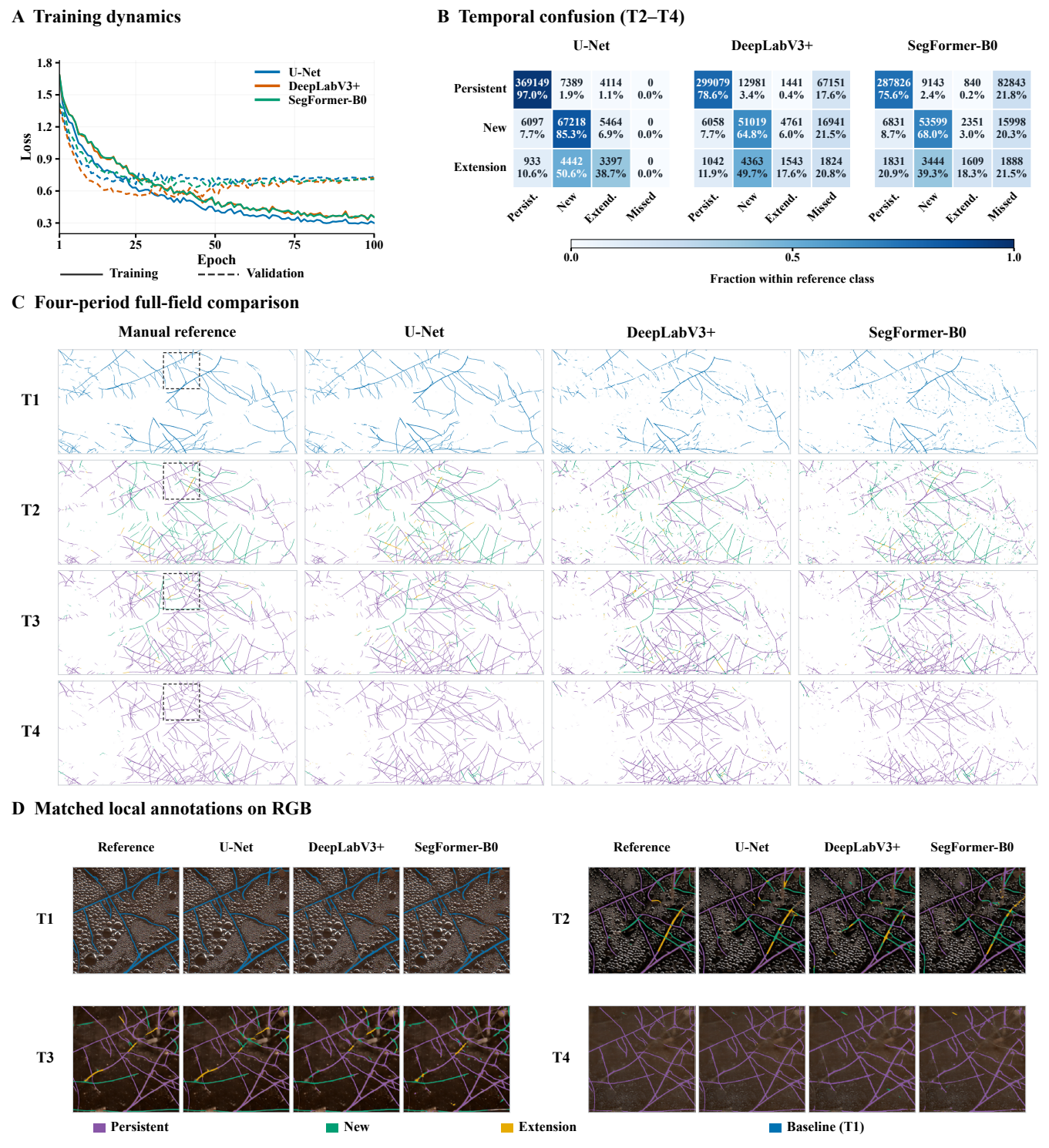

A Training dynamics
U-Net
DeepLabV3+
SegFormer-B0
Loss
Epoch
Training
Validation
B Temporal confusion (T2–T4)
U-Net
DeepLabV3+
SegFormer-B0
Persistent
New
Extension
Persist.
New
Extend.
Missed
369149 97.0%
7389 1.9%
4114 1.1%
0 0.0%
6097 7.7%
67218 85.3%
5464 6.9%
0 0.0%
933 10.6%
4442 50.6%
3397 38.7%
0 0.0%
299079 78.6%
12981 3.4%
1441 0.4%
67151 17.6%
6058 7.7%
51019 64.8%
4761 6.0%
16941 21.5%
1042 11.9%
4363 49.7%
1543 17.6%
1824 20.8%
287826 75.6%
9143 2.4%
840 0.2%
82843 21.8%
6831 8.7%
53599 68.0%
2351 3.0%
15998 20.3%
1831 20.9%
3444 39.3%
1609 18.3%
1888 21.5%
Fraction within reference class
C Four-period full-field comparison
Manual reference
U-Net
DeepLabV3+
SegFormer-B0
T1
T2
T3
T4
D Matched local annotations on RGB
Reference
U-Net
DeepLabV3+
SegFormer-B0
T1
T2
T3
T4
Persistent
New
Extension
Baseline (T1)